BookMe - Integrating an External SOAP Partner

CS543 Web Services · Assignment 3

Team ID	<TEAM-ID>
Member 1 (Team Leader)	<Roll No> - <Name>
Member 2	<Roll No> - <Name>
Member 3	<Roll No> - <Name>
Member 4	<Roll No> - <Name>

Files in this submission: integration.pdf (this document), partner.wsdl, soap-request.xml, soap-response.xml, soap-fault.xml.

1. Context (Task 1)

Partner: SecurePay Gateway - an external card-payment gateway (fictional) used by BookMe's Payment Service to collect booking payments from customers. **Operation integrated:** `ChargeBooking` (charge a customer's tokenised card for a booking).

Why SOAP on this edge? The rest of BookMe (User, Vendor, Event, Booking, Review services behind the API Gateway) is REST/JSON because those APIs are ours, change often and serve web/mobile clients. The payment edge is different: the gateway is a bank-grade partner that publishes a strict **WSDL contract**, so both sides can generate stubs and reject any malformed money message before it is processed. SOAP also gives **message-level security** (credentials and signatures travel inside the envelope Header and survive intermediaries, not just TLS hops) and mature support for **transactional/reliable messaging** - the same merchantOrderId makes a charge idempotent and auditable. Money movement is the one place BookMe wants contract-level guarantees over flexibility, so only this edge is SOAP.

Where it sits: Booking Service (REST) → Payment Service (REST, owns Payment DB) → *SOAP adapter* → SecurePay Gateway. Only the Payment Service speaks SOAP; no other BookMe service ever sees the partner's types.

2. Partner contract and messages (Tasks 2 and 3)

The WSDL is in **partner.wsdl** (document/literal, SOAP 1.1 over HTTP) and contains all six elements: *types*, *message* (ChargeBookingInput, ChargeBookingOutput, PaymentFaultMessage), *portType* (PaymentGatewayPortType / ChargeBooking), *binding* (PaymentGatewaySoapBinding), *service* and *port* (endpoint `https://soap.securepay-gateway.in/ws/payments/v1`). The three envelopes are in **soap-request.xml** (credentials in Header, call in Body), **soap-response.xml** and **soap-fault.xml**; each was checked element-for-element against the WSDL types.

WSDL item	Appears in message as
part credentials -> MerchantCredentials	soap:Header in request
part parameters -> ChargeBookingRequest	soap:Body in request
part parameters -> ChargeBookingResponse	soap:Body in success response
fault PaymentFault -> PaymentFault	soap:Fault / detail in fault response

3. HTTP binding (Task 4)

The request envelope (soap-request.xml, 934 bytes) is carried by one HTTP POST. The URL path comes from `soap:address` in service/port; SOAPAction comes from `soap:operation` in the binding.

```
POST /ws/payments/v1 HTTP/1.1
Host: soap.securepay-gateway.in
Content-Type: text/xml; charset=utf-8
Content-Length: 934
SOAPAction: "http://securepay-gateway.in/payment/v1/ChargeBooking"

<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope ...>  <!-- full body = soap-request.xml -->
  <soap:Header> <pay:MerchantCredentials> ... </pay:MerchantCredentials> </soap:Header>
  <soap:Body>   <pay:ChargeBookingRequest> ... </pay:ChargeBookingRequest> </soap:Body>
</soap:Envelope>
```

Full URL: `https://soap.securepay-gateway.in/ws/payments/v1` (HTTPS, port 443). Success returns **HTTP 200** with soap-response.xml; a SOAP Fault comes back as **HTTP 500** with soap-fault.xml, as required by SOAP 1.1 over HTTP.

4. Discovery - modern form (Task 5)

How BookMe gets the WSDL. At partner onboarding SecurePay gives BookMe a direct, versioned WSDL URL: `https://soap.securepay-gateway.in/ws/payments/v1?wsdl`. A build step downloads it, stores it in the repository as partner.wsdl (pinned, so a silent partner change cannot break production) and generates the client stub for the Payment Service. The service is then *looked up* by name from BookMe's internal service catalogue rather than hard-coding the endpoint, so staging/production endpoints can be swapped without code changes. No live UDDI server is used; the catalogue entry below follows the UDDI structure (businessEntity / businessService / bindingTemplate / tModel).

Catalogue / registry entry

```
businessEntity:
  name:        SecurePay Payments Pvt. Ltd.
  businessKey: securepay-gateway.in
  contact:     integrations@securepay-gateway.in
businessService:
  serviceKey:  securepay-card-payments
  name:        PaymentGatewayService
  category:    payments / card-charge
  version:     v1
bindingTemplate:
  accessPoint: https://soap.securepay-gateway.in/ws/payments/v1   (useType: endPoint)
  protocol:    SOAP 1.1 / HTTP, document-literal
  operations:  ChargeBooking
  soapAction:  http://securepay-gateway.in/payment/v1/ChargeBooking
tModel:                                # pointer to the contract it speaks
  tModelKey:   uddi:securepay-gateway.in:payment-v1
  name:        PaymentGatewayPortType
  overviewDoc: https://soap.securepay-gateway.in/ws/payments/v1?wsdl
  local copy:  partner.wsdl
consumer:
  name:        BookMe Payment Service
  owner:       BookMe team
```

5. Fault mapping (Task 6)

The Payment Service acts as an anti-corruption layer: it catches the partner's `soap:Fault`, reads `PaymentFault/errorCode`, and converts it to BookMe's own error vocabulary. The Booking Service and students/customers only ever see BookMe errors - never *card_declined*, gatewayReference or SOAP terms. (Assumption: BookMe's Assignment 2 `placeBooking` contract promises the errors PAYMENT_DECLINED and PAYMENT_UNAVAILABLE; adjust names if your Assignment 2 differs.)

Partner fault (faultcode / errorCode)	Meaning	BookMe error (placeBooking)	HTTP	Booking state
Client / card_declined	Issuer declined	PAYMENT_DECLINED - "Your payment was declined. Please try another payment method."	402	PAYMENT_ FAILED
Client / insufficient_funds	Not enough balance	PAYMENT_DECLINED (same message)	402	PAYMENT_ FAILED
Server / gateway_unavailable (retryable=true)	Gateway down/timeout	PAYMENT_UNAVAILABLE - "Payment is temporarily unavailable. Please try again shortly."	503	PENDING_PAYMENT (retry)
Client / invalid_credentials	Our merchant key rejected	PAYMENT_UNAVAILABLE (generic); ops alert raised internally	503	PENDING_PAYMENT
Client / duplicate_order	merchantOrderId already charged	No error - return the existing payment result (idempotent)	200	CONFIRMED

Example. The fault in soap-fault.xml (card_declined) leaves the Payment Service as this BookMe response; the partner's code and gatewayReference are logged internally only:

```
HTTP/1.1 402 Payment Required
Content-Type: application/json

{ "error": "PAYMENT_DECLINED",
  "message": "Your payment was declined. Please try another payment method.",
  "bookingId": "BKM-2026-000451" }
```

Because every partner fault collapses into one of BookMe's two published payment errors, the gateway can later be replaced (another SOAP or REST provider) without changing the REST contract that customers and the Booking Service depend on.